

MAHMOUD AHMED ABDELAWAL

AI & DATA SCIENCE ENGINEER • TECHNICAL PROJECT DOSSIER

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1. EXECUTIVE SUMMARY & DISTINCTIONS

AI and Data Science Engineer specializing in **Multimodal Intelligence**, **Computer Vision**, **Retrieval-Augmented Generation (RAG)**, and **Production MLOps**. Graduated with an Artificial Intelligence specialization from the Egyptian E-Learning University (EELU). Proven track record of winning national competitions and architecting scalable deep learning systems from research formulation to containerized FastAPI/Docker microservices.

■ **1st Place Champion — AI Hackathon:** Awarded First Place nationwide among engineering teams across Egypt with *DermaAssist AI* (WHO-grounded RAG + Multimodal Dermatology Decision Support), sponsored by the Ministry of Communications & IT, Orange Egypt, and Instant.

■ **Ministerial Recognition for Excellence:** Formally honored by the **Minister of Communications and Information Technology (MCIT) of Egypt** as one of the top-performing students nationwide in the Digital Egypt Pioneers Initiative (DEPI).

2. PROJECT 01 — S-REPORT AI: UNIVERSAL URBAN INTELLIGENCE

Category: Graduation Project (AI Architecture & Backend Lead) | **Repository:** github.com/MahmoudMahrous24/S-Report-AI-Vision

S-Report AI is a state-of-the-art hybrid multimodal computer vision and spatial intelligence platform engineered for real-time disaster detection, damage severity evaluation, and automated emergency resource dispatch.

Technical Architecture & Core Innovations:

- **Hybrid Zero-Shot Hazard Classification:** Combines OpenAI's CLIP (Contrastive Language-Image Pre-training) zero-shot visual-linguistic embeddings with a fine-tuned ConvNeXt-Tiny convolutional backbone for high-accuracy, sub-150ms classification of critical emergency situations (fire, building collapse, major flooding, vehicle accidents).
- **Spatial Scene Understanding (BLIP):** Integrates BLIP (Bootstrapping Language-Image Pre-training) to automatically synthesize rich natural language scene descriptions and situational context for first responders.
- **Monocular 3D Depth Estimation (MiDaS):** Incorporates MiDaS relative depth estimation models to construct 3D depth maps, allowing spatial hazard proximity calculations and structural collapse risk assessment.
- **Real-Time Object & Hazard Localization (YOLOv8):** Deploys YOLOv8 instance segmentation to detect, count, and outline trapped civilians, emergency vehicles, and infrastructure hazards in real time.
- **Weighted Decision & Dispatch Engine:** Engineered a mathematical multi-criteria triage algorithm that aggregates hazard severity, civilian density, and proximity into an automated priority dispatch score.
- **Production-Ready Infrastructure:** Built with an asynchronous FastAPI RESTful backend, SQLite database persistence, and Docker containerization ready for cloud deployment.

Core Stack: Python, PyTorch, CLIP, ConvNeXt, YOLOv8, BLIP, MiDaS, OpenCV, FastAPI, Docker, SQLite, NumPy

3. PROJECT 02 — DERMAASSIST AI: CLINICAL DECISION SUPPORT

Distinction: ■ 1st Place Champion — Nationwide AI Hackathon | **Repository:** github.com/MahmoudMahrous24/AI-Hackathon-Sohag

DermaAssist AI is an award-winning bilingual (Arabic/English) clinical decision support platform designed to bridge the specialist healthcare deficit in dermatology, strictly grounded in official World Health Organization (WHO) clinical guidelines.

Technical Architecture & Core Innovations:

- **WHO-Grounded RAG Pipeline:** Built an evidence-based Retrieval-Augmented Generation pipeline using ChromaDB vector database and Nomic Embeddings indexed over certified WHO dermatological clinical manuals, eliminating hallucination risks.
- **Advanced LLM Orchestration:** Integrated Google Gemini Flash via structured prompt engineering to generate accurate, bilingual diagnostic assistance, clinical rationales, and emergency warnings.
- **Multimodal Skin Lesion Analysis:** Implemented deep learning vision models capable of analyzing high-resolution dermoscopic images and live video streams for rapid lesion assessment and anomaly flagging.
- **Bilingual Voice Query Interface:** Built an intuitive voice input interface supporting both Standard Arabic, Egyptian dialect, and English to ensure accessibility for non-technical patients.
- **Dual-Portal Medical Platform:** Developed separate authenticated interfaces for patients and certified dermatologists, featuring automated prescription generation, electronic health records (EHR), and triage management.

Core Stack: React 19, TypeScript, FastAPI, Gemini Flash, ChromaDB, Nomic Embeddings, SentenceTransformers, SQLite, WebRTC

4. PROJECT 03 — ARTIFICIAL VINCENT VAN GOGH: GENERATIVE ART AI

Category: DEPI Selected Project | **Repository:** github.com/DEPI-Project/Artificial-Vincent-Van-Gogh

Artificial Vincent Van Gogh is a generative deep learning pipeline engineered to synthesize authentic artwork capturing the post-impressionist brushstroke dynamics, color palettes, and texture nuances of Vincent Van Gogh.

Technical Architecture & Core Innovations:

- **DCGANs Architecture:** Trained Deep Convolutional Generative Adversarial Networks from scratch on a curated high-resolution dataset of Van Gogh paintings, utilizing strided convolutions, batch normalization, and LeakyReLU activations.
- **Neural Style Transfer (NST):** Formulated optimization algorithms leveraging VGG-19 feature representations and Gram Matrix spatial correlations to transfer post-impressionist artistic style onto arbitrary natural photographs.
- **Loss Optimization:** Balanced perceptual loss, content loss, and style loss weights to preserve semantic structural contours while applying heavy, dynamic artistic textures.

Core Stack: Python, PyTorch, Torchvision, DCGANs, VGG-19, OpenCV, Matplotlib, NumPy, CUDA

5. PROJECT 04 — END-TO-END DATA ANALYSIS & STATISTICAL MODELING

Category: Data Science Portfolio | Repository: github.com/MahmoudMahrous24/Data-Analysis

A comprehensive collection of data analysis pipelines focused on exploratory data analysis (EDA), anomaly detection, statistical hypothesis testing, data cleaning, and actionable business insight extraction across real-world datasets.

Technical Architecture & Methodologies:

- **Exploratory Data Analysis (EDA):** Systematic profiling of dataset distributions, multivariate correlations, skewness metrics, and feature interactions.
- **Data Cleaning & Preprocessing:** Robust handling of missing data, outlier identification via IQR and Z-score methods, categorical encoding, and feature scaling.
- **Statistical Testing & Insights:** Conducting hypothesis tests (t-tests, ANOVA, Chi-Square) to extract statistically sound conclusions and guide predictive modeling.
- **Data Storytelling & Visualizations:** Generating publication-quality charts and interactive dashboards using Matplotlib, Seaborn, and Pandas.

Core Stack: Python, Pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, Seaborn, Jupyter Notebooks

6. COMPREHENSIVE SKILLS & CERTIFICATIONS MATRIX

Domain	Key Technologies & Frameworks
AI & Deep Learning	PyTorch, TensorFlow, Scikit-Learn, CNNs, Transformers, Transfer Learning, GANs, Model Optimization
Computer Vision	YOLOv8, CLIP, ConvNeXt, BLIP, MiDaS Depth Estimation, OpenCV, Image Segmentation
NLP & RAG	RAG Architectures, ChromaDB, Nomic Embeddings, Gemini Flash API, SentenceTransformers, Tokenization
Languages	Python (Expert), TypeScript, JavaScript, C++, Java, SQL
Backend & Cloud	FastAPI, Docker, Microsoft Azure, React 19, SQLite, PostgreSQL, Git, GitHub
Certifications	Microsoft ML Engineer (DEPI, 12/2025) • English & Soft Skills (DEPI, 2025) • Deep Learning, CV & NLP (2024)